SONA College of Arts and Science

Enabling Dreams Empowering Visions

Department of Computer Science

Master of Science in Computer Science

A Decision Support System for Kidney Stone Detection Using CNN

Submitted by

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SYNOPSIS

- Kidney stone detection is important for early diagnosis.
- Manual detection is time-consuming and error-prone
- Proposed system uses CNN for automatic detection
- Dataset from Kaggle (CT scan images)
- 80% training and 20% testing
- Predicts Stone / No Stone
- Web-based system using Flask

Overview

- Introduction
- Hardware Specification
- Software Specification
- Existing System
- Drawbacks of Existing System
- Proposed System
- Input Design
- Output Design
- Database Description
- Testing and Implementation
- Conclusion

CNN

Kidney Stone Detection

AI-Powered Diagnostics

INTRODUCTION

What is Kidney Stone Disease?
Kidney stones are hard deposits formed from minerals and salts in the kidney.
Early detection is important to prevent severe pain and complications.
The Challenges are:
CT scans are the gold standard for detection, but manual interpretation by radiologists is:

- Manual analysis is time-consuming
- Subject to human error and fatigue
- Depends on doctor expertise

Our Solution

- CNN Deep Learning Model
- Clinical Decision Support
- 98–99% Accuracy
- Web-Based Interface
- Auto Report Generation
- Real-Time Prediction

SYSTEM SPECIFICATION

Hardware Configuration

Processor	Intel Core i5 or equivalent
RAM	8 GB (Minimum: 4 GB)
Storage	256 GB HDD / SSD
System Type	Laptop / Desktop
Input	Keyboard & Mouse
Output	Monitor / Display Screen
OS	Windows 10 / Windows 11

Software Specification

Language	Python
Framework	Flask
Deep Learning	TensorFlow / Keras
Frontend	HTML, CSS, Bootstrap
Image Processing	OpenCV, PIL
IDE	Visual Studio Code
Report Gen	ReportLab (PDF)

SOFTWARE SPECIFICATION

- OS: Windows 10 / 11→ Platform to run application
- Language: Python→ Backend & CNN development
- Framework: Flask→ Web application handling
- Deep Learning: TensorFlow, Keras→ Model building & training
- Frontend: HTML, CSS, Bootstrap→ User interface design
- IDE: VS Code→ Coding & debugging
- Libraries: OpenCV, NumPy, PIL→ Image processingModel Format: .h5→ Stores trained model

Existing System & Drawbacks

Existing System

- Manual CT scan interpretation by radiologists
- Diagnosis depends on individual expertise
- No automated decision support mechanism
- Manual documentation and report preparation
- Inconsistent diagnostic results across clinicians

⚠️ Drawbacks

✗ High risk of human error & misdiagnosis

✗ Time-consuming & labor-intensive process

✗ Small stones may be missed due to fatigue

✗ Inconsistent results in high workload environments

✗ No confidence-based prediction support

✗ Delays in report generation & documentation

PROPOSED SYSTEM

An intelligent CNN-based Decision Support System that automates kidney stone detection from CT scan images with high accuracy and clinical usability.



CNN Model

Automatically extracts features from CT scans for accurate classification



98–99% Accuracy

High reliability deep learning model trained on kidney CT dataset



Web Interface

Flask-based user-friendly interface for patient data & image upload



Confidence Score

Prediction reliability score shown alongside diagnosis result



Auto Reports

PDF diagnostic report generation with patient details & advice



Real-Time Result

Instant prediction output with medical explanation for clinicians

DATASET DESIGN

- Dataset collected from KaggleContains CT scan images

  Includes stone and non-stone images

- Dataset divided into:

  80% Training

  20% TestingTraining

  used to learn patterns Testing used to evaluate model

INPUT DESIGN

Patient Information Form

Patient Name

Enter full name

Age

Enter age

Gender

Male / Female

Blood Pressure

e.g., 120/80

Symptoms

Describe symptoms

🔍 **Predict Kidney Stone**

CT Scan Image Upload



Upload CT Scan Image
(JPG / PNG Format)

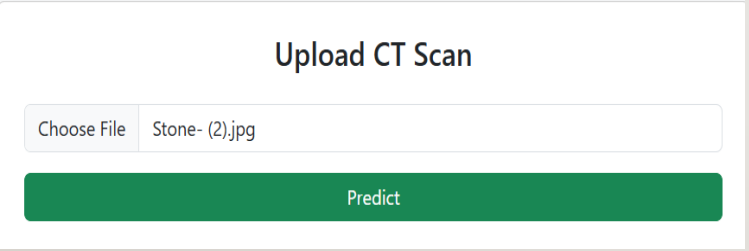
Click to Browse Files

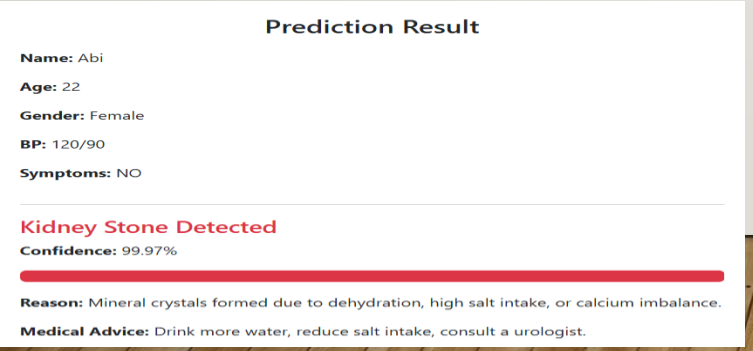
Supported Formats: JPG • PNG • JPEG

🚀 **Upload & Predict**

OUTPUT DESIGN







Medical Prediction Report

Name : Abi

Age : 22

Gender : Female

Bp : 120/90

Symptoms : NO

Prediction : Kidney Stone Detected

Confidence : 99.97000122070312%

Reason : Mineral crystals formed due to dehydration, high salt intake, or calcium imbalance.

Medical Advice : Drink more water, reduce salt intake, consult a urologist.

08 System Testing & Implementation

01

Unit Testing

Each module (patient input, image upload, preprocessing, CNN prediction) tested individually to ensure correct functionality and output.

02

Integration Testing

Verifies all modules work together seamlessly. CT image flows through preprocessing pipeline into the CNN prediction module correctly.

03

Functional Testing

Ensures all required functions — image upload, prediction, result display, and report download — perform as expected.

04

UI Testing

Web interface tested for correct rendering of forms, buttons, and result pages across different screen sizes and browsers.

✅ **CNN Model Accuracy: 98% – 99% | High Reliability & Performance Validated**

CONCLUSION

The Decision Support System for Kidney Stone Detection Using CNN has been successfully designed and developed. The system integrates deep learning with a practical web application to deliver a reliable, accurate, and user-friendly diagnostic support tool.

- 🎯 CNN model achieves 98–99% accuracy on kidney CT dataset
- ⚡ Automated detection reduces manual radiologist workload
- 📊 Confidence score improves clinical trust in the system
- 🌐 Flask web app enables easy access and patient data entry
- 📄 Automated PDF reports support documentation and record-keeping
- 🏥 Practical medical AI tool applicable in clinical environments

"AI-assisted diagnostics — improving outcomes, one scan at a time."

THANK YOU

A Decision Support System for Kidney Stone Detection Using CNN

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